

Low-Rank Matrix and Tensor-Based Reconstruction

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9.1 Introduction

Low-rank modeling [1] is a powerful approach to reconstruct images from sparsely sampled $(\mathbf{k}, t)$ -space data. Like compressed sensing, low-rank methods leverage efficient representations of an image to reduce its degrees of freedom and allow its recovery from sparse samples. A major difference is that, in low-rank reconstructions, the “sparse domain” is adaptive, changing scan to scan based on correlations observed within the acquired data.

Low-rank image reconstruction is used primarily in the context of recovering dynamic images, multicontrast images (e.g., for parameter mapping), multichannel images, and/or image patch arrays. A collection of multiple images or patches naturally lends itself to an interpretation of image sequences and images as arrays, expressed either as matrices (2D arrays) or generalizable to higher-order tensors (≥ 3 D arrays)—especially when imaging multiple simultaneous dynamic processes. Spatiotemporal correlation induces linear dependence between columns, rows, and fibers of the array, leading to low-rankness which is then exploited for accelerated imaging.

A wide range of imaging applications has been shown to admit low-rank representations, including cardiovascular imaging [2–5], quantitative parameter mapping and multicontrast imaging [6–10], MR fingerprinting [11–13], dynamic contrast enhancement imaging [1,14,15], spectroscopic imaging [16–18], and fMRI [19–21], among others. Furthermore, the use of higher-order tensor models has also allowed simultaneous imaging of various combinations of such dynamic processes [22–26].

Section 9.2 generalizes dynamic imaging and multicontrast imaging problems in particular to a context that will permit high-order tensor interpretations; Section 9.3 introduces several low-rank matrix models; Section 9.4 extends modeling to higher-order tensors; and Section 9.5 concludes the chapter with a brief outlook.

9.2 Problem formulation

This chapter focuses on reconstructing dynamic images and multicontrast images (sequences of multiple image frames changing with, e.g., motion or variable pulse sequence/timing parameters). We generalize the concept of dynamic and multicontrast imaging by first considering an underlying multi-dimensional image $\tilde{x}(\mathbf{r}, \tau_1(t), \tau_2(t), \dots, \tau_L(t))$, which is a function of voxel location $\mathbf{r} = [x, y, z]^T$ and $L \geq 1$ time-varying independent variables $\{\tau_\ell(t)\}_{\ell=1}^L$, each representing a different physical or physiological dynamic process, including but not limited to motion, NMR relaxation, and dynamic contrast

enhancement. Progression of these dynamic processes changes the image that can be observed as well as the signal that can be measured.

The dynamic MR signal from $\tilde{x}$ is measured in $(\mathbf{k}, t)$ -space:

$$s_c(\mathbf{k}, t) = E_c \{x(\mathbf{r}, t)\} + \eta_c(\mathbf{k}, t) \quad (9.1)$$

$$x(\mathbf{r}, t) = \tilde{x}(\mathbf{r}, \tau_1(t), \tau_2(t), \dots, \tau_L(t)), \quad (9.2)$$

where $E_c\{\cdot\}$ spatially encodes x at coil c and η denotes complex Gaussian noise. The image $x(\mathbf{r}, t)$ describes how $\tilde{x}$ plays out in the scanner as the $\{\tau_\ell(t)\}_{\ell=1}^L$ vary.

It is common to target only a single dynamic process by “freezing” all others during acquisition, i.e., the ℓ th dynamic process represented by $\tau_\ell(t)$ is often isolated by acquiring data only when $\tau_q(t) = \kappa_q \ \forall q \neq \ell$, for some set of constant values $\{\kappa_q\}_{q \neq \ell}$. This is practically achieved by: 1) pausing the actual processes, physically holding the $\tau_q(t)$ constant (e.g., pausing respiration with breath holds); and/or 2) synchronizing acquisition to the process, only collecting or storing $(\mathbf{k}, t)$ -space data acquired when $\tau_q(t)$ matches κ_q (e.g., removing the influence of cardiac motion with electrocardiogram (ECG) triggering to a specific cardiac phase). The decision of which dynamic processes to retain and which to remove depends on the specific imaging application.

The goal of low-rank image reconstruction is to recover either x or an $\tilde{x}(\mathbf{r}, \tau_\ell)$ with a single dynamic process (Sections 9.3 and 9.4.2) or the entire $\tilde{x}(\mathbf{r}, \tau_1, \tau_2, \dots, \tau_L)$ (Section 9.4.3) from undersampled $(\mathbf{k}, t)$ -space measurements through the use of low-rank image modeling.

9.3 Matrix-based approaches

This section focuses on reconstructing $x(\mathbf{r}, t)$ using low-rank matrix models.

It is useful to begin with the singular value decomposition (SVD), which decomposes any matrix $\mathbf{X} \in \mathbb{C}^{M \times N}$ into a sum of rank-1 matrices:

$$\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H = \sum_{r=1}^{\min(M, N)} \sigma_r \mathbf{u}_r \mathbf{v}_r^H. \quad (9.3)$$

The left and right complex singular vectors $\{\mathbf{u}_r\}_{r=1}^{\min(M, N)}$ and $\{\mathbf{v}_r\}_{r=1}^{\min(M, N)}$ are the columns of $\mathbf{U}$ and $\mathbf{V}$, and are orthonormal bases for the columns of $\mathbf{X}$ and $\mathbf{X}^H$, respectively. The singular values $\{\sigma_r\}_{r=1}^{\min(M, N)}$ lie along the diagonal of $\mathbf{\Sigma}$ ($[\mathbf{\Sigma}]_{rr} = \sigma_r$), and are real nonnegative values in descending order. The rank of $\mathbf{X}$ is equal to the number of nonzero singular values; when there are only $R < \min(M, N)$ nonzero singular values (i.e., when $\sigma_r = 0 \ \forall r > R$), then $\text{rank}(\mathbf{X}) = R < \min(M, N)$, and $\mathbf{X}$ is said to be low-rank. In this case, singular values and vectors corresponding to $r > R$ are unnecessary for signal representation. For a rank- R $\mathbf{X}$, the summation in Eq. (9.3) can therefore be truncated without loss of information, such that $\mathbf{X}$ is the sum of R rank-1 matrices: $\mathbf{X} = \sum_{r=1}^R \sigma_r \mathbf{u}_r \mathbf{v}_r^H$. The ability to efficiently represent a low-rank matrix by its truncated SVD implies that low-rank matrices have fewer degrees of freedom than full-rank matrices: Whereas a full-rank $\mathbf{X} \in \mathbb{C}^{M \times N}$ has $2MN$ real degrees of freedom, a rank- R $\mathbf{X}$ has only $2(M + N - R)R$ real degrees of freedom. Note also that truncating the SVD below the true rank of $\mathbf{X}$ ($R < \text{rank}(\mathbf{X})$) produces the least-squares optimal rank- R approximation

to $\mathbf{X}$. As a result, the SVD is a critical component of several low-rank reconstruction and approximation algorithms.

A more general decomposition of $\mathbf{X} \in \mathbb{C}^{M \times N}$ is the matrix factorization

$$\mathbf{X} = \mathbf{A}\Phi. \quad (9.4)$$

Here, the left matrix $\mathbf{A} \in \mathbb{C}^{M \times R}$ and right matrix $\Phi \in \mathbb{C}^{R \times N}$ are constrained neither to be orthonormal, nor to have any special ordering of columns and rows, respectively. As a result, the decomposition in Eq. (9.4) is not unique. The SVD can provide forms of this factorization, for example, by calculating $\mathbf{A} = \mathbf{U}\Sigma$, $\Phi = \mathbf{V}^H$, but it is only one example construction. Note that the decomposition in Eq. (9.4) places an upper bound on the rank of $\mathbf{X}$, specifically that $\text{rank}(\mathbf{X}) \leq R$ (ranks lower than R occur if either the columns of $\mathbf{A}$ or the rows of Φ are linearly dependent). Its generality is useful for low-rank reconstruction algorithms and will be used in this chapter to tie together several low-rank image models.

To reconstruct $x(\mathbf{r}, t)$ using low-rank matrix models, we can express Eq. (9.1) in matrix-vector form as $\mathbf{s} = E(\mathbf{X}) + \eta$, where the matrix $\mathbf{X} \in \mathbb{C}^{M \times N}$ has elements $[\mathbf{X}]_{mn} = x(\mathbf{r}_m, t_n)$, and $E(\cdot)$ models spatial encoding:

$$\mathbf{X} = \begin{bmatrix} x(\mathbf{r}_1, t_1) & x(\mathbf{r}_1, t_2) & \cdots & x(\mathbf{r}_1, t_N) \\ x(\mathbf{r}_2, t_1) & x(\mathbf{r}_2, t_2) & \cdots & x(\mathbf{r}_2, t_N) \\ \vdots & \vdots & \ddots & \vdots \\ x(\mathbf{r}_M, t_1) & x(\mathbf{r}_M, t_2) & \cdots & x(\mathbf{r}_M, t_N) \end{bmatrix}.$$

The matrix $\mathbf{X}$ is sometimes referred to as the Casorati matrix of $x(\mathbf{r}, t)$. Note that a Casorati matrix can also be formed from $\tilde{x}(\mathbf{r}, \tau_\ell)$ with a single dynamic process as $[\mathbf{X}]_{mn} = \tilde{x}(\mathbf{r}_m, \tau_{\ell,n})$. As a result, the approaches discussed here also apply to imaging an $\tilde{x}(\mathbf{r}, \tau_\ell)$, for which $(\mathbf{k}, t)$ -space data may have already been reorganized according to, e.g., the cardiac phase.

9.3.1 Global low-rank modeling

The R th-order partial separability (PS) model [1] expresses $x(\mathbf{r}, t)$ as

$$x(\mathbf{r}, t) = \sum_{r=1}^R a_r(\mathbf{r})\varphi_r(t), \quad (9.5)$$

decomposing the image into R spatial coefficient maps $\{a_r(\mathbf{r})\}_{r=1}^R$ and R temporal functions $\{\varphi_r(t)\}_{r=1}^R$ (Fig. 9.1). This model is consistent with the matrix factorization in Eq. (9.4), where $\mathbf{A}$ has elements $[\mathbf{A}]_{mr} = a_r(\mathbf{r}_m)$ and Φ has elements $[\Phi]_{rn} = \varphi_r(t_n)$. This image property is termed *partial* separability because it separates groups of variables (e.g., $\mathbf{r}$ as a whole) as opposed to every individual variable (e.g., each x, y, z).

This formulation reveals that strong spatiotemporal correlation in $x(\mathbf{r}, t)$ induces $\mathbf{X}$ to be low-rank: When the set of voxel functions $\{x(\mathbf{r}_m, t)\}_{m=1}^M$ (the rows of $\mathbf{X}$) and the set of image frames $\{x(\mathbf{r}, t_n)\}_{n=1}^N$ (the columns of $\mathbf{X}$) are each so highly correlated to be linearly dependent, $\mathbf{X}$ will be low-rank such that $\text{rank}(\mathbf{X}) < \min(M, N)$. A low model order $R < \min(M, N)$ provides an avenue for image reconstruction from sparse $(\mathbf{k}, t)$ -space samples due to the reduced degrees of freedom in low-rank matrices described previously. See Fig. 9.2.

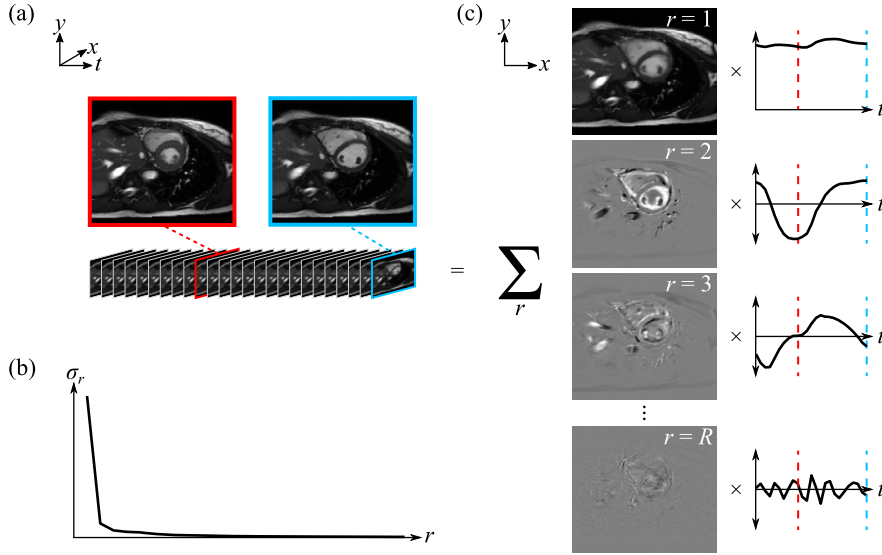


FIGURE 9.1 Global low-rank decomposition

(a) A cardiac MR cine image sequence with 25 frames. (b) Singular value plot demonstrating the quick decay and low effective rank of the image matrix. (c) Decomposition of the image into spatial coefficient maps and their corresponding temporal functions.

Eq. (9.4) implies that $\mathbf{X}$ lies in either of two R -dimensional subspaces: the column space of $\mathbf{A}$ and the row space of Φ . In the context of Eq. (9.5), the column space of $\mathbf{A}$ becomes a spatial subspace, and the row space of Φ becomes a temporal subspace. Similarly, when the columns of $\mathbf{A}$ or the rows of Φ are orthonormal, they constitute a spatial basis or a temporal basis for $\mathbf{X}$, respectively. This subspace view can inform data acquisition and image reconstruction (as will be seen in Sections 9.3.1.1 and 9.3.1.2, respectively) and reveals a connection to the synthesis formulation of compressed sensing (CS) with a temporal sparsifying basis. In CS, Φ is a pre-selected sparsifying basis with $R = N$, and $\mathbf{A}$ are the sparse coefficients in the transform domain $\mathbf{A} = \mathbf{X}\Phi^{-1}$. Therefore, where CS seeks a sparse $\mathbf{A}$ with $R = N$ and pre-selected Φ , low-rank image reconstruction seeks an adaptive “sparsifying basis” Φ that depends on the measured signal, and also seeks an $\mathbf{A}$ which is not necessarily sparse. In the context of low-rank image models, “sparsity” is instead achieved by rank-reduction, which induces the sparsity of the singular values: A rank- R matrix has only R nonzero singular values (i.e., $\sigma_r = 0 \forall r > R$). The fast decay of singular values for an example dynamic MR image is displayed in Fig. 9.1b.

9.3.1.1 Sampling

When low-rank image reconstruction is to be employed, there are two primary considerations for sampling $(\mathbf{k}, t)$ -space: 1) incoherence and 2) whether to collect interleaved “training data”.

As discussed in Chapter 8, measurement incoherence is dependent on both the sampling pattern and the sparsifying basis used in image reconstruction. For low-rank image reconstruction, there would ideally be incoherence between the sampling operator and the low-rank “sparsifying basis” Φ [27].

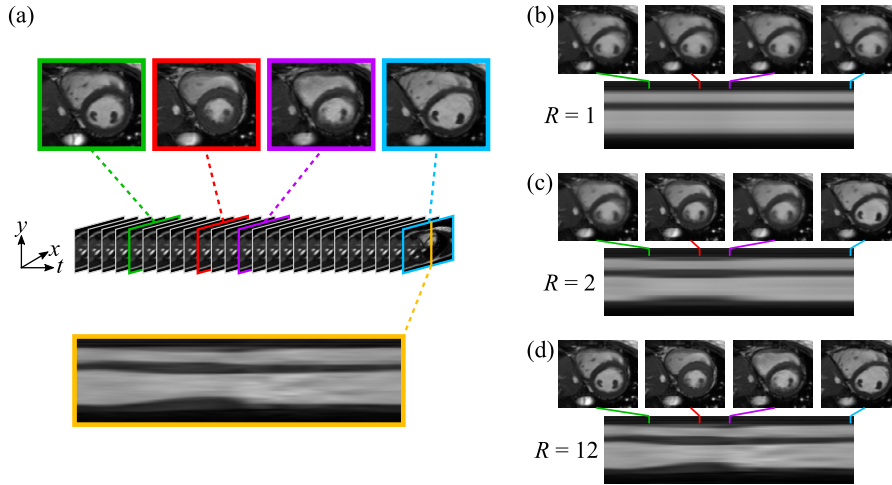


FIGURE 9.2 Effect of rank on image approximation

(a) Zoomed-in regions of four frames and a temporal profile of the cardiac cine MR image, and its (b) rank-1, (c) rank-2, and (d) rank-12 SVD approximations. Note that morphology does not change in a rank-1 image since there is only one spatial coefficient map being modulated over time. Additional dynamic detail is present as rank increases, and the $R = 12$ image is virtually indistinguishable from the original 25-frame dynamic image.

Since Φ is adaptive in low-rank image modeling, it is generally not known *a priori* exactly how much incoherence a given sampling pattern will achieve, presenting a challenge for optimal sampling design. However, some known-coherent scenarios can be readily avoided. For example, a repetitive sampling pattern synchronized to a periodic dynamic $\tau_\ell(t)$ (e.g., cardiac motion, respiratory motion, magnetization preparation intervals) would result in strong coherence between the sampling operator and Φ . This coherence can be mitigated by avoiding sampling pattern periodicity within the physiologically expected ranges for cardiac and respiratory motion and synchronization with the known magnetization preparation period. Incoherence can also be practically achieved in most scenarios through randomly ordered, nonrepetitive sampling patterns—a common choice when sampling for low-rank image reconstruction. Real-time motion tracking techniques such as phase ordering with automatic window selection (PAWS) [28] can update sampling patterns on the fly to minimize redundant samples, which can also benefit incoherence. However, all of these are only heuristic approaches for achieving a good sampling patterns; none of them directly measure or optimize incoherence between the sampling operator and Φ , as it is not known *a priori* in the presence of motion. Real-time feedback approaches have been proposed to estimate Φ while data are still being acquired and subsequently update sampling patterns in real-time [29], but have not yet been adopted in practice. In situations where Φ can be estimated *a priori* (e.g., for multicontrast imaging/parameter mapping without motion [6,30]), incoherence can be measured and optimized prospectively.

The second consideration is the inclusion of regular frequent sampling of a subset of $\mathbf{k}$ -space, variably referred to as training data, subspace training data, navigator data, dense data, or auxiliary data.

These sampling patterns interleave training data alongside incoherent sparse sampling of $(\mathbf{k}, t)$ -space (sometimes referred to as imaging data or sparse data) at more $\mathbf{k}$ -space locations. The rationale for this is two-fold: 1) because low-rank models decompose $\mathbf{X}$ into a spatial factor $\mathbf{A}$ and a temporal factor Φ , the classical trade-off between spatial and temporal resolution is somewhat decoupled, and sampling can be split into subsets—one targeting $\mathbf{A}$ and the other targeting Φ . The imaging data have the extensive $\mathbf{k}$ -space coverage required for high spatial resolution and therefore target $\mathbf{A}$; the training data are sampled very frequently and therefore target Φ . 2) Interleaved acquisition also makes two-step explicit recovery of $\mathbf{A}$ and Φ possible since Φ can be directly estimated from the training data, after which image reconstruction reduces to recovering $\mathbf{A}$ from the imaging data given a known Φ . However, collecting training data usually comes at the cost of collecting more imaging data elsewhere in $\mathbf{k}$ -space; if every N th readout is a training readout instead of an imaging readout, then the imaging data acquisition efficiency is $\frac{N-1}{N}$ (e.g., 50% efficient when collecting training data every other readout, and 87.5% efficient when collecting training data every eighth readout). Furthermore, frequent training readouts also have the potential to induce eddy-current artifacts from large jumps in $\mathbf{k}$ -space, especially for balanced SSFP sequences [31]. Although there are self-contained training data schemes that collect training data during slice rephase and read dephase gradients that already occur prior to imaging readouts [32], this strategy extends the minimum achievable echo time and is therefore primarily appropriate for T_2^* -weighted imaging.

9.3.1.2 Image reconstruction

There are two basic approaches to image reconstruction: 1) explicit low-rank reconstruction, which aims to recover a rank- R $\mathbf{X}$ (typically by recovering the factors $\mathbf{A}$ and Φ); and 2) implicit low-rank reconstruction, which aims to recover an $\mathbf{X}$ which minimizes a rank surrogate.

Explicit low-rank reconstruction

The objective of explicit low-rank reconstruction is to reconstruct a $\mathbf{X}$ with at most rank R . This sets up a rank-constrained optimization problem

$$\hat{\mathbf{X}} = \arg \min_{\mathbf{X}} \|\mathbf{E}(\mathbf{X}) - \mathbf{s}\|_2^2 \quad s.t. \quad \text{rank}(\mathbf{X}) \leq R, \quad (9.6)$$

which can be parameterized as

$$\{\hat{\mathbf{A}}, \hat{\Phi}\} = \arg \min_{\mathbf{A}, \Phi} \|\mathbf{E}(\mathbf{A}\Phi) - \mathbf{s}\|_2^2. \quad (9.7)$$

Eq. (9.7) frames the problem as recovery of the model factors $\mathbf{A}$ and Φ from measured data. This permits efficient image reconstruction algorithms that store and operate upon $\mathbf{A}$ and Φ instead of storing and operating upon the larger $\mathbf{X}$ matrix; in fact, $\mathbf{X}$ needs never to be explicitly calculated [33], permitting image reconstruction even on systems that cannot hold the full $\mathbf{X}$ in memory.

Fixed-subspace reconstruction

Several imaging methods use a two-step approach to recover $\mathbf{A}$ and Φ [1,3]. These methods require collecting training data as previously described in Section 9.3.1.1 that can then be arranged into a

multichannel $(\mathbf{k}, t)$ -space Casorati matrix $\mathbf{S}_{\text{tr}}$ that lies in the row space of Φ :

$$\mathbf{S}_{\text{tr}} = \begin{bmatrix} \mathbf{S}_{\text{tr},1} \\ \mathbf{S}_{\text{tr},2} \\ \vdots \\ \mathbf{S}_{\text{tr},C} \end{bmatrix}, \quad \mathbf{S}_{\text{tr},c} = \begin{bmatrix} s_c(\mathbf{k}_1, t_1) & s_c(\mathbf{k}_1, t_2) & \cdots & s_c(\mathbf{k}_1, t_N) \\ s_c(\mathbf{k}_2, t_1) & s_c(\mathbf{k}_2, t_2) & \cdots & s_c(\mathbf{k}_2, t_N) \\ \vdots & \vdots & \ddots & \vdots \\ s_c(\mathbf{k}_M, t_1) & s_c(\mathbf{k}_M, t_2) & \cdots & s_c(\mathbf{k}_M, t_N) \end{bmatrix}.$$

The two image reconstruction steps are then serial recovery of Φ and $\mathbf{A}$ from the training data and imaging data, respectively:

$$\hat{\Phi} = \arg \min_{\Phi} \left\| \mathbf{S}_{\text{tr}} - \mathbf{S}_{\text{tr}} \left(\Phi^\dagger \Phi \right) \right\|_F^2, \quad (9.8)$$

$$\hat{\mathbf{A}} = \arg \min_{\mathbf{A}} \left\| E(\mathbf{A} \hat{\Phi}) - \mathbf{s} \right\|_2^2. \quad (9.9)$$

Eq. (9.8) can be efficiently solved by calculating the SVD of $\mathbf{S}_{\text{tr}}$ and defining $\hat{\Phi}$ from the R most significant right singular vectors. With a known $\hat{\Phi}$, image reconstruction then reduces to Eq. (9.9), simple linear least-squares recovery of $\mathbf{A}$. Eq. (9.9) can be viewed as recovering the coordinates of $\mathbf{X}$ within a fixed subspace (the row space of $\hat{\Phi}$).

Although fixed-subspace reconstruction is computationally efficient, it does of course require collection of training data, which comes at the cost of sparser sampling elsewhere in $\mathbf{k}$ -space, as described in Section 9.3.1.1. Two-step reconstruction also potentially introduces model bias through its choice of $\hat{\Phi}$: Although the temporal basis calculated from the SVD of the training data is least squares optimal for describing $\mathbf{S}_{\text{tr}}$, it is generally not the optimal temporal basis for describing $\mathbf{X}$ as well [32], unless certain conditions on the training $\mathbf{k}$ -space trajectory as well as the spark and joint sparsity of the columns of $\mathbf{X}$ are satisfied [34]. Notably, this strategy requires advance selection of the model order R , typically by heuristically fixing R for a given application or by analyzing the singular values of $\mathbf{S}_{\text{tr}}$.

Alternating reconstruction

To address this, schemes for jointly recovering $\mathbf{A}$ and Φ from the whole of the collected data have been proposed [35,36]. These methods seek a solution to Eq. (9.7) for both $\mathbf{A}$ and Φ . This optimization problem is nonconvex, but a satisfactory local minimum can typically be found by alternating least squares (ALS) minimization:

$$\hat{\Phi}^{[it]} = \arg \min_{\Phi} \left\| E \left(\hat{\mathbf{A}}^{[it-1]} \Phi \right) - \mathbf{s} \right\|_2^2 \quad (9.10)$$

$$\hat{\mathbf{A}}^{[it]} = \arg \min_{\mathbf{A}} \left\| E \left(\mathbf{A} \hat{\Phi}^{[it]} \right) - \mathbf{s} \right\|_2^2, \quad (9.11)$$

each iteration of which updates Φ and $\mathbf{A}$ by linear least squares recovery. Fast convergence can be achieved by starting from $R = 1$ and periodically incrementing R during optimization until the desired model order is reached [37]. This approach does not require any training data, but as a nonconvex optimization technique is dependent on initial guesses. Like fixed-subspace reconstruction, this approach also requires selection of the specific rank R .

Implicit low-rank reconstruction

The objective of implicit low-rank reconstruction is to recover a matrix $\mathbf{X}$ without explicitly choosing an R or parameterizing $\mathbf{X}$ as $\mathbf{A}\Phi$. Without a known R , low-rank image reconstruction can be reframed as a rank-minimization problem

$$\hat{\mathbf{X}} = \arg \min_{\mathbf{X}} \text{rank}(\mathbf{X}) \quad \text{s.t.} \quad \|E(\mathbf{X}) - \mathbf{s}\|_2^2 \leq \epsilon \quad (9.12)$$

that seeks the lowest-rank $\mathbf{X}$ within a predetermined data consistency tolerance ϵ . Eq. (9.12) is nonconvex and can be NP-hard, so several alternative rank surrogates have been proposed in the framework of regularized reconstruction:

$$\hat{\mathbf{X}} = \arg \min_{\mathbf{X}} \|E(\mathbf{X}) - \mathbf{s}\|_2^2 + \lambda R_{\text{rank}}(\mathbf{X}), \quad (9.13)$$

where $R_{\text{rank}}(\cdot)$ is a rank surrogate and λ is the regularization hyperparameter that controls effective rank.

A natural surrogate for matrix rank uses the Schatten p -norm $\|\cdot\|_p$ [4,38]

$$R_{\text{rank}}(\mathbf{X}) = \|\mathbf{X}\|_p^p = \sum_{r=1}^{\min(M,N)} [\sigma_r(\mathbf{X})]^p, \quad (9.14)$$

that promotes sparsity of the singular values of $\mathbf{X}$ when $p \leq 1$. The Schatten p -norm for $p < 1$ is still nonconvex, so a common practical choice is $p = 1$ [4,20,39], which corresponds to the nuclear norm $\|\cdot\|_*$, the tightest convex relaxation of matrix rank [40,41]:

$$R_{\text{rank}}(\mathbf{X}) = \|\mathbf{X}\|_* = \sum_{r=1}^{\min(M,N)} \sigma_r(\mathbf{X}). \quad (9.15)$$

Because the nuclear norm is analogous to the L1-norm of the singular values, Eq. (9.15) can, for example, be solved by iterative soft-thresholding of the singular values, known as the singular value thresholding algorithm [42].

A major benefit of implicit low-rank matrix-recovery approaches that solve forms of Eq. (9.13) is that they do not require *a priori* knowledge of matrix rank R , but they do still require determination of a hyperparameter λ to control the matrix effective rank. Like explicit ALS approaches, implicit approaches do not require collection of training data, allowing flexibility in sampling. However, these benefits come at the cost of computational efficiency: By recovering a $\mathbf{X}$ that can be factored rather than by recovering the factors $\mathbf{A}$ and Φ , the memory usage of implicit low-rank reconstruction algorithms typically do not benefit from low-rank matrix compression to the same extent as explicit low-rank reconstruction algorithms. Furthermore, when $\mathbf{X}$ is not parameterized, Fourier encoding and sensitivity encoding must be calculated for all N image frames of $\mathbf{X}$ during each iteration, rather than just for the R frames of $\mathbf{A}$. For small N , these computational considerations are not generally a concern, but can become prohibitive for large-scale dynamic image recovery.

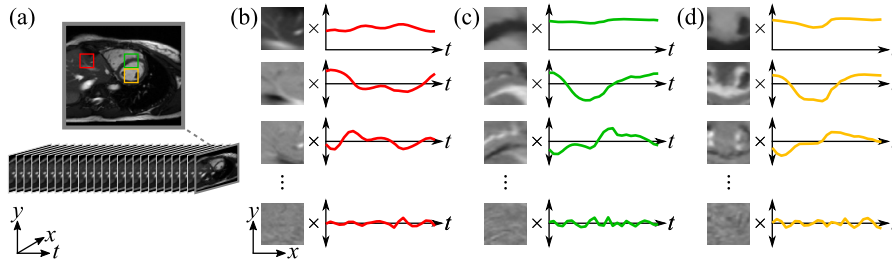


FIGURE 9.3 Local low-rank decomposition

(a) The dynamic image can be split into several spatial regions or patches. (b, c, d) Each patch constitutes its own low-rank matrix and has its own temporal basis that does not have to be shared with other patches. Note that the patch in the liver (*red* (gray in print version)) has quite different temporal basis functions than the two distant patches over the heart (*green*, *yellow* (mid gray, light gray in print version)), which neighbor each other and have more similar temporal basis functions.

9.3.2 Local low-rank modeling

The global low-rank model in Eq. (9.5) assumes that all voxels are globally correlated, i.e., that the entire set of voxel functions $\{x(\mathbf{r}_m, t)\}_{m=1}^M$ lies in the same temporal subspace $\text{span}\{\varphi_r(t)\}_{r=1}^R$. However, correlation may be higher in neighboring voxels from the same organ than in distant voxels from other organs, in which case it is more efficient to restrict the assumption of spatiotemporal correlation to local voxel neighborhoods (e.g., image patches) via local low-rank (LLR) modeling [43,7,10,44,45].

LLR models split the image into multiple spatial regions $\{\Omega_i\}_{i \geq 1}$, each of which is assigned to its own temporal subspace $\text{span}\{\varphi_{i,r}(t)\}_{r=1}^{R_i}$:

$$x(\mathbf{r}, t) = \begin{cases} \sum_{r=1}^{R_1} a_r(\mathbf{r}) \varphi_{1,r}(t), & \text{if } \mathbf{r} \in \Omega_1 \\ \sum_{r=1}^{R_2} a_r(\mathbf{r}) \varphi_{2,r}(t), & \text{if } \mathbf{r} \in \Omega_2 \\ \vdots \\ \sum_{r=1}^{R_i} a_r(\mathbf{r}) \varphi_{i,r}(t), & \text{if } \mathbf{r} \in \Omega_i \\ \vdots \end{cases} \quad (9.16)$$

This model allows different patches of $x(\mathbf{r}, t)$ to lie in different temporal subspaces (Fig. 9.3). The image $x(\mathbf{r}, t)$ can be said to lie in a union of subspaces rather than a single shared temporal subspace.

The LLR property can be imposed during image reconstruction by solving

$$\hat{\mathbf{X}} = \arg \min_{\mathbf{X}} \|\mathbf{E}(\mathbf{X}) - \mathbf{s}\|_2^2 + \lambda \sum_{i \geq 1} R_{\text{rank}}(\mathbf{P}_i \mathbf{X}), \quad (9.17)$$

where $\mathbf{P}_i$ extracts the i th spatial patch/region (a subset of the rows of $\mathbf{X}$) to form a spatially-localized Casorati matrix. This implicit LLR formulation does not require strict adherence to Eq. (9.16), in the sense that it optionally permits the use of overlapping patches to reduce artifacts at the boundaries of

patches. LLR image reconstruction is generally compatible with explicit ALS algorithms. However, it is not always compatible with fixed-subspace modeling since it is difficult to determine a separate Φ_i for each spatial region/patch. Still, fixed-subspace reconstruction is sometimes possible when the training scheme allows low spatial resolution reconstruction of the training data [46,47] or when the regions are allowed to share some temporal basis functions but vary in model order [48].

When there is more local correlation than global correlation, LLR reconstruction can produce less temporal blurring from model bias than global low-rank reconstruction, at the expense of ignoring any global correlation that may still be present. LLR reconstruction especially benefits when the average patch model order can be set quite low, in which case the additional degrees of freedom associated with allowing multiple temporal subspaces are counteracted by the lower average model order. LLR approaches require selection of a patch size: Small patch sizes typically have very low rank, but may not be sensitive enough to capture all nearby temporally correlated structures; on the other hand, large patch sizes capture more correlation but are not very specific, and may have higher rank due to the inclusion of more distant, less temporally correlated structures. A patch size that spans the entire image field of view is equivalent to the global low-rank model.

9.3.3 Low-rank and sparse modeling

It is important to note that compressed sensing with fixed sparsifying bases and low-rank image modeling with adaptive bases are not mutually exclusive. For both techniques, incoherence is commonly achieved by the same means (e.g., randomized sampling), so their data acquisition requirements are often compatible. As Chapter 8 has established, MR images are known to be sparse, and as Sections 9.3.1 and 9.3.2 have established, several types of MR images are also known to be globally or locally low-rank. As a result, several methods have been developed to restrict the solution space to those images that are both low-rank and sparse (L&S).

In practice, low-rank image reconstruction can be combined with compressed sensing by introducing a sparse regularization term $\|\Psi(\mathbf{A}\Phi)\|_1$ (for explicit low-rank reconstruction) or $\|\Psi(\mathbf{X})\|_1$ (for implicit low-rank reconstruction) to the main objective function, where $\Psi(\cdot)$ is a spatial and/or temporal sparsifying transform. The L&S forms of the previous optimization problems Eqs. (9.7), (9.9), (9.13), and (9.17) become:

$$\{\hat{\mathbf{A}}, \hat{\Phi}\} = \arg \min_{\mathbf{A}, \Phi} \|E(\mathbf{A}\Phi) - \mathbf{s}\|_2^2 + \lambda \|\Psi(\mathbf{A}\Phi)\|_1 \quad (9.18)$$

$$\hat{\mathbf{A}} = \arg \min_{\mathbf{A}} \|E(\mathbf{A}\hat{\Phi}) - \mathbf{s}\|_2^2 + \lambda \|\Psi(\mathbf{A}\hat{\Phi})\|_1 \quad (9.19)$$

$$\hat{\mathbf{X}} = \arg \min_{\mathbf{X}} \|E(\mathbf{X}) - \mathbf{s}\|_2^2 + \lambda_1 R_{\text{rank}}(\mathbf{X}) + \lambda_2 \|\Psi(\mathbf{X})\|_1 \quad (9.20)$$

$$\hat{\mathbf{X}} = \arg \min_{\mathbf{X}} \|E(\mathbf{X}) - \mathbf{s}\|_2^2 + \lambda_1 \sum_{i \geq 1} R_{\text{rank}}(\mathbf{P}_i \mathbf{X}) + \lambda_2 \|\Psi(\mathbf{X})\|_1, \quad (9.21)$$

where the choice of Ψ is motivated by the specific MRI application. This can be a fixed transform [4,33] to impose transform sparsity on $\mathbf{X}$ in direct analogy to compressed sensing; in the cases of Eqs. (9.18) and (9.19), Ψ can be an adaptive transform incorporating $\Phi^\dagger$ (e.g., $\Psi(\mathbf{X}) = \Psi\mathbf{X}\Phi^\dagger = \Psi\mathbf{A}$) to impose transform sparsity on the spatial coefficients $\mathbf{A}$ and to retain the memory benefits of explicit low-rank modeling [36,48,49]. The additional regularization power of compressed sensing allows more flexible

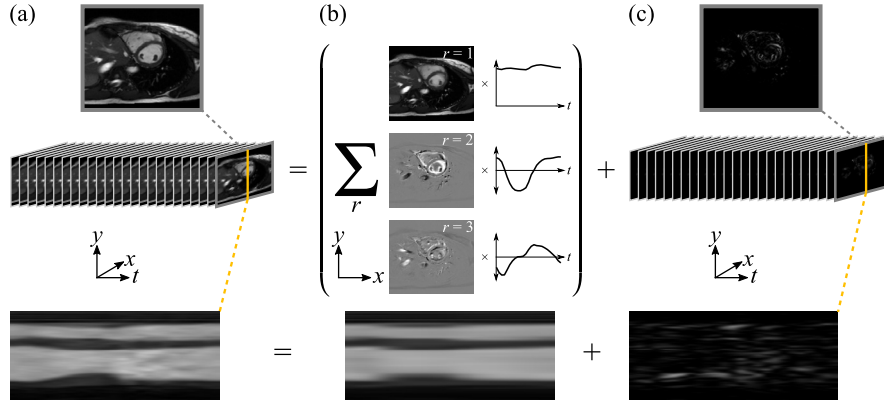


FIGURE 9.4 Low-rank plus sparse decomposition

(a) A dynamic image can also be decomposed into the sum of a (b) low-rank term and (c) a sparse (or transform sparse) term. The low-rank term can be set with a very low model order, leaving additional detail to be absorbed into the sparse term. Note here that the sparse term is being displayed as a magnitude image at 3x brightness for clarity.

selection of low-rank model orders since sparse regularization improves the conditioning of inverse problems with high R or low λ_1 [33].

9.3.4 Low-rank plus sparse modeling

An alternative approach to incorporating sparse modeling into low-rank image reconstruction is to model the image not as low-rank *and* sparse, but rather as a low-rank term *plus* a sparse residual [50–53]. The use of a low-rank plus sparse (L+S) decomposition $\mathbf{X} = \mathbf{L} + \mathbf{S}$ uses a low-rank matrix $\mathbf{L}$ to coarsely capture correlation, absorbing finer detail and transient events into the sparse residual $\mathbf{S}$ (Fig. 9.4). This decomposition is unique as long as $\mathbf{L}$ is not sparse and $\mathbf{S}$ is not low-rank [52].

L+S modeling can be used with either the explicit parameterization $\mathbf{L} = \mathbf{A}\Phi$ [53] or with implicit low-rank modeling penalizing a rank surrogate of $\mathbf{L}$ [50–52]. Explicit low-rank modeling of $\mathbf{L}$ suggests the formulation

$$\{\hat{\mathbf{A}}, \hat{\Phi}, \hat{\mathbf{S}}\} = \arg \min_{\mathbf{A}, \Phi, \mathbf{S}} \|E(\mathbf{A}\Phi + \mathbf{S}) - \mathbf{s}\|_2^2 + \lambda \|\text{vec}(\mathbf{S})\|_1, \quad (9.22)$$

where $\|\text{vec}(\mathbf{S})\|_1$ promotes sparsity of $\mathbf{S}$. Either the fixed-subspace reconstruction or ALS strategies could be applied to solve Eq. (9.22). Note that, although parameterization of $\mathbf{L} = \mathbf{A}\Phi$ has memory benefits, $\mathbf{S}$ is still stored in full.

Likewise, implicit low-rank modeling of $\mathbf{L}$ suggests the formulation

$$\{\hat{\mathbf{L}}, \hat{\mathbf{S}}\} = \arg \min_{\mathbf{L}, \mathbf{S}} \|E(\mathbf{L} + \mathbf{S}) - \mathbf{s}\|_2^2 + \lambda_1 R_{\text{rank}}(\mathbf{L}) + \lambda_2 \|\text{vec}(\mathbf{S})\|_1, \quad (9.23)$$

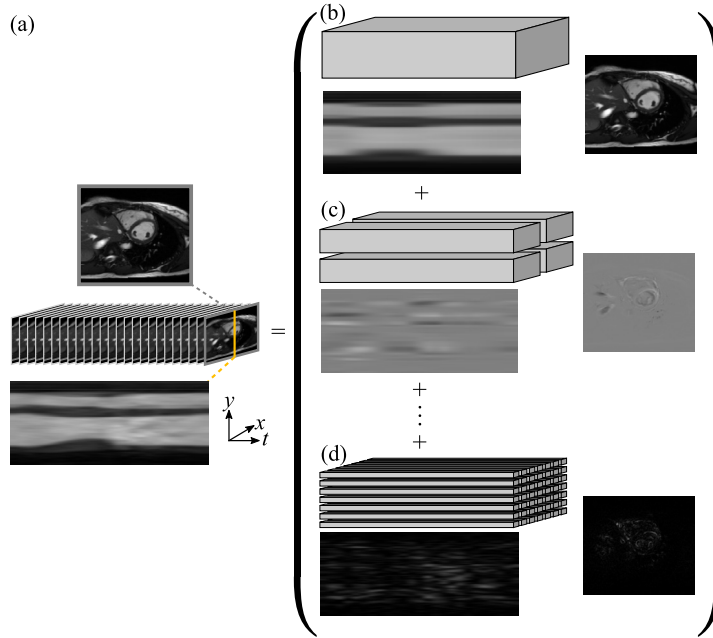


FIGURE 9.5 Multiscale low-rank decomposition

(a) A dynamic image can also be represented as the sum of low-rank images at several scales: (a) global, (b) local, (c) voxel-wise. Note that the voxel-wise scale can be treated as a sparse term, displayed here as a magnitude image at 3x brightness.

where $R_{\text{rank}}(\mathbf{L})$ promotes low-rankness of $\mathbf{L}$. This formulation has roots in robust principal component analysis (RPCA) [54] but as applied to matrix recovery rather than image denoising or analysis.

In the event that $\mathbf{S}$ is sparser in a different domain, both Eqs. (9.22) and (9.23) can be modified to impose transform sparsity of $\mathbf{S}$, simply by replacing the regularization term $\|\text{vec}(\mathbf{S})\|_1$ with a transform sparse regularization term $\|\Psi(\mathbf{S})\|_1$.

9.3.5 Multiscale low-rank modeling

Multiscale low-rank modeling generalizes the L+S concept by further decomposing the $\mathbf{L}$ term as the sum of globally low-rank and locally low-rank components, thereby bridging global low-rank modeling, local low-rank modeling, and L+S modeling using a single framework [55,56]. The multiscale low-rank decomposition expresses the image matrix as $\mathbf{X} = \sum_{j=1}^J \mathbf{X}_j$, where $\mathbf{X}_j$ is the term at the j th spatial scale: $\mathbf{X}_1$ is globally low rank, $\mathbf{X}_2$ through $\mathbf{X}_{J-1}$ are locally low-rank for increasingly smaller patch sizes, and the voxel-wise term $\mathbf{X}_J$ is analogous to the sparse term in L+S modeling (Fig. 9.5). Hence, this decomposition generalizes global low-rank modeling ($\mathbf{X}_1$ only), local low-rank modeling ($\mathbf{X}_j$ only, $1 < j < J$), and L+S modeling ($\mathbf{X}_1$ and $\mathbf{X}_J$ only).

Multiscale low-rank modeling is compatible with both explicit low-rank factorization [56] and implicit low-rank regularization [55]. The explicit formulation expresses each term as $\mathbf{X}_j = \mathcal{M}_j(\mathbf{A}_j, \{\Phi_{ij}\}_{i \geq 1})$, where $\mathbf{A}_j \in \mathbb{C}^{M \times R_j}$ are spatial coefficients as previously described, where $\Phi_{ij} \in \mathbb{C}^{R_j \times N}$ is the temporal factor for the i th spatial patch at the j th spatial scale, and where $\mathcal{M}_j$ selectively multiplies each patch of $\mathbf{A}_j$ by the corresponding Φ_{ij} to produce an $M \times N$ matrix. Image reconstruction can then be carried out in the basic framework of

$$\arg \min_{\{\mathbf{A}_j\}_{j \geq 1}, \{\Phi_{ij}\}_{i \geq 1, j \geq 1}} \left\| E \left(\sum_{j=1}^J \mathcal{M}_j(\mathbf{A}_j, \{\Phi_{ij}\}_{i \geq 1}) \right) - \mathbf{s} \right\|_2^2. \quad (9.24)$$

In practice, Eq. (9.24) incorporates an additional regularization term to further penalize the rank at each spatial scale: $\sum_j \lambda_j (\|\mathbf{A}_j\|_F^2 + \sum_i \|\Phi_{ij}\|_F^2)$, which draws upon a relaxation of the nuclear norm for explicitly parameterized low-rank matrices [56].

An implicit multiscale low-rank formulation is

$$\arg \min_{\mathbf{X}} \left\| E \left(\sum_{j=1}^J \mathbf{X}_j \right) - \mathbf{s} \right\|_2^2 + \sum_{i \geq 1} \sum_{j=1}^J \lambda_j R_{\text{rank}}(\mathbf{P}_{ij} \mathbf{X}_j), \quad (9.25)$$

where $\mathbf{P}_{ij}$ extracts the i th spatial patch at the j th spatial scale. Note that the storage requirements for the whole set of unfactored $\{\mathbf{X}_j\}_{j=1}^J$ can be prohibitive for large-scale problems, motivating the explicit strategy [56] when imaging at ambitious scales.

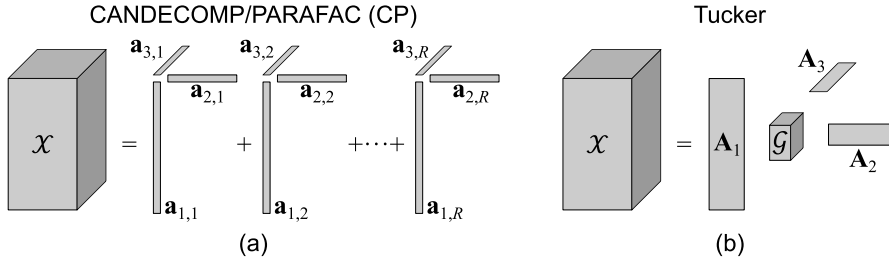
9.4 Tensor-based approaches

Whereas Section 9.3 interpreted image sequences as matrices (2-dimensional arrays), this section interprets them as higher-order tensors (≥ 3 -dimensional arrays). There are several ways to define additional dimensions that permit a tensor interpretation, e.g., coil dimensions from multichannel receive arrays, preservation of separate spatial dimensions, aggregation of similar patches from different spatial locations, and recovery of the multidynamic $\tilde{x}(\mathbf{r}, \tau_1, \tau_2, \dots, \tau_L)$ with multiple “time dimensions” $\{\tau_\ell\}_\ell^L$.

9.4.1 Tensor definitions

Tensor notation, operations, and decompositions are not as universally defined as matrix operations so here we introduce some general background before moving on to tensor-based imaging.

We refer to an N -dimensional array $\mathcal{X}$ as an N -way tensor $\mathcal{X} \in \mathbb{C}^{M_1 \times M_2 \times \dots \times M_N}$. The collection of values with all indices fixed except along the n th tensor mode (or dimension) is referred to as the mode- n fiber; fibers are simply the higher-order analogues of matrix columns (mode-1 fibers) and rows (mode-2 fibers). When extracted from tensors, fibers are typically represented as column vectors, regardless of original mode. It is sometimes useful to unfold or flatten tensors into matrices. The mode- n unfolding of $\mathcal{X}$ into a matrix is denoted as $\mathbf{X}_{(n)} \in \mathbb{C}^{M_n \times \prod_{i \neq n} M_i}$, which places the n th dimension of $\mathcal{X}$ along the columns of $\mathbf{X}_{(n)}$. Like matrices, tensors can be multiplied. We denote the n -mode matrix product of $\mathcal{X}$ with a matrix $\mathbf{Q} \in \mathbb{C}^{J \times M_n}$ as $\mathcal{Y} = \mathcal{X} \times_n \mathbf{Q}$ with $\mathcal{Y} \in \mathbb{C}^{M_1 \times \dots \times M_{n-1} \times J \times M_{n+1} \times \dots \times M_N}$, which has


FIGURE 9.6 Conceptual illustration of tensor decompositions

Decomposition of a 3-way tensor $\mathcal{X}$ in (a) CP form and (b) Tucker form.

elements $[\mathcal{Y}]_{m_1 m_2 \dots m_{n-1} j m_{n+1} \dots m_N} = \sum_{m_n=1}^{M_n} [\mathbf{Q}]_{j m_n} [\mathcal{X}]_{m_1 m_2 \dots m_{n-1} m_n m_{n+1} \dots m_N}$. This can be compactly represented in mode- n unfolded form as $\mathbf{Y}_{(n)} = \mathbf{Q} \mathbf{X}_{(n)}$.

Similarly to low-rank matrix image reconstruction, low-rank tensor (LRT) image reconstruction can also be divided into explicitly low-rank methods (that parameterize $\mathcal{X}$ by decomposing it into several factors) and implicitly low-rank methods (that minimize a surrogate for the rank of $\mathcal{X}$). A great many tensor decompositions are available for this parameterization and for the definition of tensor rank [57], but we will focus on the two most commonly used for MRI: canonical decomposition [58]/parallel factors [59] (CANDECOMP/PARAFAC, further abbreviated as CP), and the Tucker decomposition [60] (Fig. 9.6). The interested reader is referred to [57] for in-depth discussion of these and other tensor decompositions.

9.4.1.1 CP decomposition

The CP decomposition expresses $\mathcal{X}$ as the sum of rank-1 tensors:

$$\mathcal{X} = \sum_{r=1}^R \mathbf{a}_{1,r} \circ \mathbf{a}_{2,r} \circ \dots \circ \mathbf{a}_{N,r}, \quad (9.26)$$

where $\circ$ denotes the vector outer product and $\mathbf{a}_{n,r} \in \mathbb{C}^{M_n \times 1}$. Element-wise, this becomes

$$[\mathcal{X}]_{m_1 m_2 \dots m_N} = \sum_{r=1}^R [\mathbf{a}_{1,r}]_{m_1} [\mathbf{a}_{2,r}]_{m_2} \dots [\mathbf{a}_{N,r}]_{m_N} \quad (9.27)$$

$$= \sum_{r=1}^R [\mathbf{A}_1]_{m_1 r} [\mathbf{A}_2]_{m_2 r} \dots [\mathbf{A}_N]_{m_N r}, \quad (9.28)$$

where $\mathbf{A}_n \in \mathbb{C}^{M_n \times R}$ is the factor matrix for the n th dimension. This further admits the mode-1 unfolded representation

$$\mathbf{X}_{(1)} = \mathbf{A}_1 (\mathbf{A}_N \odot \mathbf{A}_{N-1} \odot \dots \odot \mathbf{A}_2)^T, \quad (9.29)$$

where $\odot$ denotes the Khatri–Rao product. Using the CP decomposition, $\mathcal{X}$ is said to be rank- R .

9.4.1.2 Tucker decomposition

The Tucker decomposition decomposes $\mathcal{X}$ with a different model order for each dimension:

$$\mathcal{X} = \sum_{r_1=1}^{R_1} \sum_{r_2=1}^{R_2} \cdots \sum_{r_N=1}^{R_N} g_{r_1 r_2 \cdots r_N} \mathbf{a}_{1,r_1} \circ \mathbf{a}_{2,r_2} \circ \cdots \circ \mathbf{a}_{N,r_N}. \quad (9.30)$$

Element-wise, this becomes

$$[\mathcal{X}]_{m_1 m_2 \cdots m_N} = \sum_{r_1=1}^{R_1} \sum_{r_2=1}^{R_2} \cdots \sum_{r_N=1}^{R_N} g_{r_1 r_2 \cdots r_N} [\mathbf{a}_{1,r_1}]_{m_1} [\mathbf{a}_{2,r_2}]_{m_2} \cdots [\mathbf{a}_{N,r_N}]_{m_N} \quad (9.31)$$

$$= \sum_{r_1=1}^{R_1} \sum_{r_2=1}^{R_2} \cdots \sum_{r_N=1}^{R_N} g_{r_1 r_2 \cdots r_N} [\mathbf{A}_1]_{m_1 r_1} [\mathbf{A}_2]_{m_2 r_2} \cdots [\mathbf{A}_N]_{m_N r_N}, \quad (9.32)$$

where $\mathbf{A}_n \in \mathbb{C}^{M_n \times R_n}$ is the factor matrix for the n th dimension. Typically, the columns within each $\mathbf{A}_n$ are chosen to be orthonormal, such that $\mathbf{A}_n$ constitutes a basis for the n th dimension of $\mathcal{X}$. This admits the n -mode product representation

$$\mathcal{X} = \mathcal{G} \times_1 \mathbf{A}_1 \times_2 \mathbf{A}_2 \cdots \times_N \mathbf{A}_N \quad (9.33)$$

that gives rise to a core tensor $\mathcal{G} \in \mathbb{C}^{R_1 \times R_2 \times \cdots \times R_N}$ with elements $[\mathcal{G}]_{r_1 r_2 \cdots r_N} = g_{r_1 r_2 \cdots r_N}$. This core tensor governs the interactions between the factor matrices and is the higher-order analogue to the matrix singular values but does not have to be diagonal [61]. The mode-1 unfolded representation becomes

$$\mathbf{X}_{(1)} = \mathbf{A}_1 \mathbf{G}_{(1)} (\mathbf{A}_N \otimes \mathbf{A}_{N-1} \otimes \cdots \otimes \mathbf{A}_2)^T, \quad (9.34)$$

where $\otimes$ denotes the Kronecker product. Using the Tucker decomposition, $\mathcal{X}$ is said to be rank- $(R_1, R_2, \dots, R_N)$.

The CP and Tucker decompositions have both been used for a wide range of applications, with CP tending to be used for analysis and Tucker tending to be used for compression (in many scenarios, each R_n for the Tucker decomposition is often much smaller than R from the CP decomposition, leading to smaller factor matrices) [57]. In the context of MRI, the CP decomposition may permit more accurate signal representation [26], at the expense of memory usage. For matrices (i.e., $N = 2$), both the CP and Tucker decompositions lead to the same representation.

9.4.1.3 Tensor rank surrogates

As in low-rank matrix image reconstruction, implicit low-rank tensor reconstruction will seek an $\mathcal{X}$ which minimizes a surrogate for tensor rank. As the comparison of rank definitions for the CP and Tucker decompositions shows, there are several definitions of tensor rank, and the task of determining the rank of a known tensor can be NP-hard [62]. As such, there are also several surrogates $R_{\text{rank}}(\cdot)$ for tensor rank [63–65], the most well-known of which is the sum of nuclear norms of each unfolding of $\mathcal{X}$ [63]:

$$R_{\text{rank}}(\mathcal{X}) = \sum_{n=1}^N \|\mathbf{X}_{(n)}\|_*, \quad (9.35)$$

which serves as a convex relaxation of tensor rank. However, the formulations in this section extend to other definitions of $R_{\text{rank}}(\mathcal{X})$ as well.

9.4.2 Reinterpreting dynamic images as tensors

A dynamic or multicontrast image $x(\mathbf{r}, t)$ or $\tilde{x}(\mathbf{r}, \tau_\ell)$ is not limited to a matrix formation as in Section 9.3; several higher-dimensional formations are possible as well. Additional dimensions can be defined from multiple receive coil channels, by enforcing separability of the individual spatial dimensions, by stacking similar image patches centered at different voxels, and more.

9.4.2.1 Coil modeling

Modern MR systems use multiple receive channels for signal detection, for which the image $x_c(\mathbf{r}, t)$ received at channel c varies with the coil sensitivity pattern $x_c(\mathbf{r}, t) = C_c(\mathbf{r})x(\mathbf{r}, t)$. When stacking the images from different receive channels into a tensor $\mathcal{X} \in \mathbb{C}^{M \times N \times C}$ with elements $[X]_{mnc} = x_c(\mathbf{r}_m, t_n)$, coil compressibility and smoothness of the coil sensitivity patterns $\{C_c(\mathbf{r})\}_{c=1}^C$ will induce weak global correlation and strong local correlation in $\mathcal{X}$ [66,67]. As a result, LRT modeling can be used to perform calibrationless parallel dynamic imaging without knowledge of the coil sensitivities or an autocalibration signal. The best-known form of this reconstruction problem takes the form of an implicit LLR tensor problem [68]:

$$\arg \min_{\mathbf{X}} \|E(\mathcal{X}) - \mathbf{s}\|_2^2 + \lambda \sum_{i \geq 1} R_{\text{rank}}(\mathcal{X} \times_1 \mathbf{P}_i), \quad (9.36)$$

where as in Eq. (9.17), $\mathbf{P}_i$ extracts the i th spatial patch, but where $E(\cdot)$ no longer includes multiplication by known coil sensitivities. This problem is also compatible with explicit ALS formulations.

9.4.2.2 Patch similarity modeling

Additional dimensionality can also be defined by stacking spatially correlated patches from various spatial locations [69–73]. Several sets of patches can be aggregated and then formed into a tensor, either by stacking correlated static patches from individual timepoints (subsets of single columns of $\mathbf{X}$) into a matrix or by stacking correlated dynamic patches (N -column submatrices of $\mathbf{X}$) into a tensor. We will denote the j th collection of patches as $\mathcal{X}_j$. Enforcing low-rankness of each $\mathcal{X}_j$ promotes spatial or spatiotemporal self-similarity throughout $x(\mathbf{r}, t)$.

Low-rankness of the $\{\mathcal{X}_j\}_{j \geq 1}$ can be enforced by solving a problem such as

$$\arg \min_{\mathbf{X}} \|E(\mathbf{X}) - \mathbf{s}\|_2^2 + \lambda \sum_{j \geq 1} R_{\text{rank}}(P_j(\mathbf{X})), \quad (9.37)$$

where P_j concatenates a set of several similar patches from $\mathbf{X}$ into a matrix or tensor such that $\mathcal{X}_j = P_j(\mathbf{X})$.

There are also several ways to systematically stack shifting, overlapping spatial or $\mathbf{k}$ -space patches that fall into the class of structured low-rank matrix completion methods, as will be discussed in Chapter 10.

9.4.2.3 Spatial separability

The methods discussed so far have flattened spatial dimensions, conceptualizing an image frame as a vector that is placed as a column of $\mathbf{X}$. Several methods opt to instead preserve spatial dimensionality without vectorization [74]. An $M_x \times M_y \times M_z \times N$ image sequence is therefore taken to be a $\mathcal{X} \in \mathbb{C}^{M_x \times M_y \times M_z \times N}$ with elements $[\mathcal{X}]_{k\ell mn} = x([x_k, y_\ell, z_m]^T, t_n)$.

Whereas Eq. (9.5) assumes low-order partial separability of $\mathbf{X}$ by grouping together all voxels, LRT treatments of this $\mathcal{X}$ assume low-order separability of all variables:

$$x(\mathbf{r}, t) = \sum_{r=1}^R u_r(x) v_r(y) w_r(z) \varphi_r(t) \quad (9.38)$$

or

$$x(\mathbf{r}, t) = \sum_{d=1}^D \sum_{p=1}^P \sum_{q=1}^Q \sum_{r=1}^R g_{dpqr} u_d(x) v_p(y) w_q(z) \varphi_r(t), \quad (9.39)$$

corresponding to the CP and Tucker decompositions, respectively. These decompositions place stronger assumptions on $x(\mathbf{r}, t)$ than Eq. (9.5) since they assume not only spatiotemporal correlation, but also low-rank spatial structure (linearly-dependent rows/columns/fibers).

Implicit low-rank reconstruction of $\mathcal{X}$ [74,75] can be performed by solving:

$$\arg \min_{\mathbf{X}} \|E(\mathcal{X}) - \mathbf{s}\|_2^2 + \lambda R_{\text{rank}}(\mathcal{X}). \quad (9.40)$$

Separable imaging can also be performed with explicit low-rank modeling by parameterizing $\mathcal{X}$ according to, e.g., Eqs. (9.38) and (9.39) [76–78]. Furthermore, the many variations of low-rank modeling described in Section 9.3 can be readily extended to accommodate spatial separability, including LLR [26,79], L&S, L+S [80], and multiscale LRT image reconstruction.

9.4.3 Multidynamic tensors

Sections 9.3 and 9.4.2 previously discussed reconstructing $x(\mathbf{r}, t)$ or an $\tilde{x}(\mathbf{r}, \tau_\ell)$, which required isolating a single dynamic process. Here we instead discuss reconstructing the underlying multidynamic image $\tilde{x}(\mathbf{r}, \tau_1, \tau_2, \dots, \tau_L)$, which has L “time dimensions” corresponding to the independent variables $\{\tau_\ell\}_{\ell=1}^L$ [81,22,82,24]. This multidimensional structure lends itself well to tensor representations, as $\tilde{x}$ can be readily expressed as an $(L+1)$ -way tensor $\mathcal{X} \in \mathbb{C}^{M \times N_1 \times N_2 \times \dots \times N_L}$ with elements $[\mathcal{X}]_{mn_1 n_2 \dots n_L} = \tilde{x}(\mathbf{r}_m, \tau_{1, n_1}, \tau_{2, n_2}, \dots, \tau_{L, n_L})$.

There are several benefits of multidynamic imaging of $\tilde{x}(\mathbf{r}, \tau_1, \tau_2, \dots, \tau_L)$ over imaging $x(\mathbf{r}, t)$ or an $\tilde{x}(\mathbf{r}, \tau_\ell)$ with a single time dimension. For one, multidynamic imaging does not require dynamic processes to be frozen by prospectively enforcing $\tau_q(t_n) = \kappa_q \forall q \neq \ell, n = 1, 2, \dots, N$. Instead, all L dynamic processes are resolved during imaging, after which an equivalent “frozen” image can be retroactively produced from $\tilde{x}$ by extracting the temporal slice at $\tau_q = \kappa_q \forall q \neq \ell: \tilde{x}(\mathbf{r}, \kappa_1, \kappa_2, \dots, \tau_\ell, \dots, \kappa_L)$. This reduces dependence on techniques such as breath holds and ECG triggering since motion can be resolved in $\tilde{x}$ instead of paused or removed. Furthermore, by allowing $\{\kappa_q\}_{q \neq \ell}$ to be chosen retrospectively, imaging decisions such as contrast weighting can be made after imaging rather than scouting for,

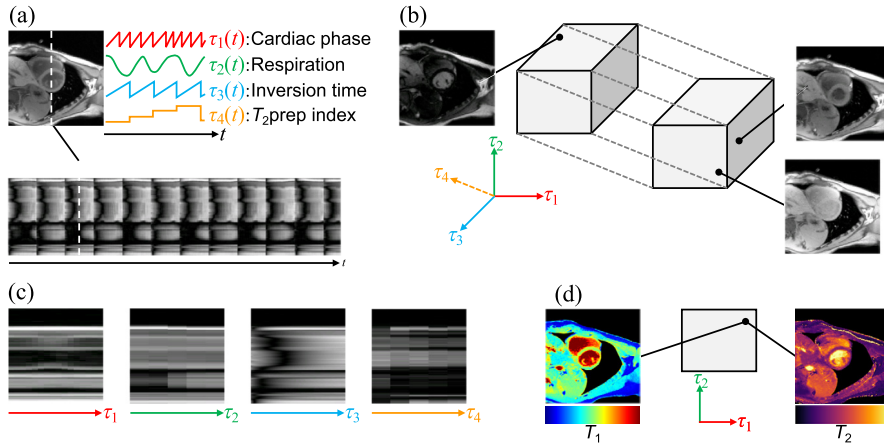


FIGURE 9.7 Illustration of the multidynamic imaging concept

(a) Imaging $L = 4$ overlapping dynamic processes: cardiac motion, respiratory motion, T1 recovery and T2 decay. (b) Four-dimensional hypercube representing all four time dimensions. Tensor reconstruction [24] enables (c) visualization of retrospectively isolated dynamic processes, and (d) quantitative maps at each motion state combination. Source: Adapted with permission from Ref. [83].

e.g., ideal inversion times for nulling tissue. Multidynamic imaging is especially beneficial when multiple dynamic processes are of clinical interest (e.g., different types of motion in addition to multiple contrast weightings for parametric mapping) because recovering the entire $\tilde{x}(\mathbf{r}, \tau_1, \tau_2, \dots, \tau_L)$ in one scan eliminates the need to perform multiple scans, each resolving a different individual τ_ℓ (Fig. 9.7).

9.4.3.1 Tensor-based compressed sensing

High-dimensional CS can be used to reconstruct $\mathbf{X}$ by solving a problem such as

$$\hat{\mathcal{X}} = \arg \min_{\mathcal{X}} \|E(\mathcal{X}) - \mathbf{s}\|_2^2 + \sum_{i=1}^{L+1} \lambda_i \|\text{vec}(\Psi_i \mathbf{X}_{(i)})\|_1, \quad (9.41)$$

where the rows of $\mathbf{X}_{(1)} \in \mathbb{C}^{M \times \prod_{\ell=1}^L N_\ell}$ index $\mathbf{r}$, the rows of $\mathbf{X}_{(i)} \in \mathbb{C}^{N_{i-1} \times M \times \prod_{\ell \neq i-1} N_\ell}$ index τ_{i-1} for $i > 1$, and the sparsifying transform Ψ_i operates along $\mathbf{r}$ for $i = 1$ and τ_{i-1} for $i > 1$.

The advantage of multidynamic CS over single-dynamic CS—in addition to the general benefits of multidynamic imaging—is that an image can typically be more efficiently compressed along each individual time dimension τ_ℓ rather than along t . For example, the difference image along τ_ℓ represents the change from a single dynamic process

$$\nabla_{\tau_\ell} \{\tilde{x}\} = \tilde{x}(\mathbf{r}, \tau_1, \tau_2, \dots, \tau_\ell, \dots, \tau_L) - \tilde{x}(\mathbf{r}, \tau_1, \tau_2, \dots, \tau_\ell - \Delta\tau_\ell, \dots, \tau_L), \quad (9.42)$$

which can be sparser than the difference between successive timepoints

$$\nabla_t \{x\} = x(\mathbf{r}, t) - x(\mathbf{r}, t - \Delta t)$$

$$= \tilde{x}(\mathbf{r}, \tau_1(t), \tau_2(t), \dots, \tau_L(t)) - \tilde{x}(\mathbf{r}, \tau_1(t - \Delta t), \tau_2(t - \Delta t), \dots, \tau_L(t - \Delta t)), \quad (9.43)$$

which represents changes from multiple processes.

The XD-GRASP method [81] introduced high-dimensional compressed sensing for dynamic MRI, using $L = 2$ for cardiac- and respiratory-resolved imaging of the heart and for respiratory- and DCE-resolved abdominal imaging. The XD flow method [82] performed high-dimensional compressed sensing with different $L = 3$ combinations of cardiac, respiratory, flow and DCE time dimensions, collapsing the fourth dimension not included in each combination.

9.4.3.2 Multidynamic low-rank tensor modeling

A complementary alternative to CS is low-rank tensor image modeling. Revisiting Eq. (9.5) in the context of multidynamic imaging, we have:

$$\tilde{x}(\mathbf{r}, \tau_1, \tau_2, \dots, \tau_L) = \sum_{r=1}^R a_r(\mathbf{r}) \tilde{\varphi}_r(\tau_1, \tau_2, \dots, \tau_L), \quad (9.44)$$

which simply replaces the $\{\varphi_r(t)\}_{r=1}^R$ with multidimensional functions $\{\tilde{\varphi}_r(\tau_1, \tau_2, \dots, \tau_L)\}_{r=1}^R$. In discretized form, Eq. (9.44) becomes

$$\mathcal{X} = \Phi \times_1 \mathbf{A}, \quad (9.45)$$

where $\Phi \in \mathbb{C}^{R \times N_1 \times \dots \times N_L}$ has elements $[\Phi]_{rn_1n_2\dots n_L} = \tilde{\varphi}_r(\tau_{1,n_1}, \tau_{2,n_2}, \dots, \tau_{L,n_L})$.

LRT modeling assumes low-order separability of each $\tilde{\varphi}_r(\tau_1, \tau_2, \dots, \tau_L)$. More specifically, the CP decomposition of $\mathbf{X}$ implies

$$\tilde{\varphi}_r(\tau_1, \tau_2, \dots, \tau_L) = \varphi_{1,r}(\tau_1) \varphi_{2,r}(\tau_2) \dots \varphi_{N,r}(\tau_L), \quad (9.46)$$

where $\{\varphi_{\ell,r}(\tau_\ell)\}_{r=1}^R$ spans the subspace for the ℓ th time dimension. The Tucker decomposition of $\mathcal{X}$ implies

$$\tilde{\varphi}_r(\tau_1, \tau_2, \dots, \tau_L) = \sum_{r_1=1}^{R_1} \sum_{r_2=1}^{R_2} \dots \sum_{r_L=1}^{R_L} g_{rr_1r_2\dots r_L} \varphi_{1,r_1}(\tau_1) \varphi_{2,r_2}(\tau_2) \dots \varphi_{N,r_L}(\tau_L), \quad (9.47)$$

where $\{\varphi_{\ell,r_\ell}(\tau_\ell)\}_{r_\ell=1}^{R_\ell}$ spans the subspace for the ℓ th time dimension. Fig. 9.8 depicts the Tucker decomposition of a cardiac image with $L = 3$ time dimensions: cardiac motion, respiration, and inversion recovery.

Explicit multidynamic low-rank tensor reconstruction

Explicit low-rank tensor image reconstruction is possible by parameterizing $\mathcal{X}$ according to Eq. (9.45) and either Eq. (9.46) or Eq. (9.47). Image reconstruction can for example proceed according to the CP recovery problem

$$\left\{ \hat{\mathbf{A}}, \left\{ \hat{\Phi}_\ell \right\}_{\ell=1}^L \right\} = \arg \min_{\mathbf{A}, \{\Phi_\ell\}_{\ell=1}^L} \left\| E \left(\sum_{r=1}^R \mathbf{a}_r \circ \Phi_{1,r} \circ \Phi_{2,r} \circ \dots \circ \Phi_{L,r} \right) - \mathbf{s} \right\|_2^2, \quad (9.48)$$

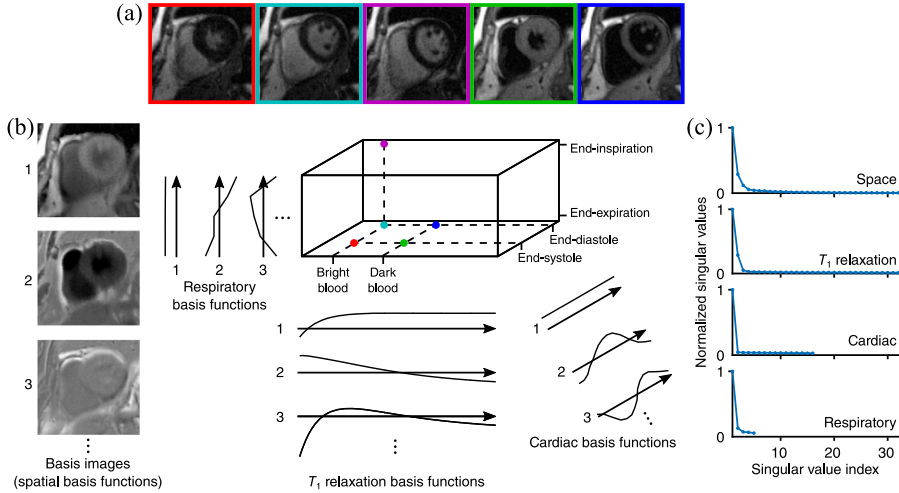


FIGURE 9.8 Illustration and analysis of multiple time dimensions for multidynamic imaging

(a) Locations of different images in a space with $L = 3$ time dimensions. Different T_1 weightings lie along the inversion time axis (*horizontal*), different cardiac phases lie along the cardiac time axis (*depth*), and different respiratory phases lie along the respiratory time axis (*vertical*). (b) The three most significant orthogonal basis functions in Tucker form, as reconstructed from 1-min worth of data using MR Multitasking [24]. (c) Singular value curves from the higher-order SVD of 12.3-min worth of raw training data (i.e., enough data to cover all motion-state and contrast combinations), demonstrating that the singular values decay quickly for all unfoldings of the raw data tensor. *Source: Adapted with permission from Ref. [24].*

where the elements of $\mathbf{a}_r \in \mathbb{C}^{M \times 1}$ are $[\mathbf{a}_r]_m = [\mathbf{A}]_{mr}$ and the elements of $\varphi_{\ell,r} \in \mathbb{C}^{N_\ell \times 1}$ are $[\varphi_{\ell,r}]_n = \varphi_{\ell,r}(\tau_{\ell,n})$, or according to the Tucker recovery problem

$$\left\{ \hat{\mathcal{G}}, \hat{\mathbf{A}}, \left\{ \hat{\Phi}_\ell \right\}_{\ell=1}^L \right\} = \arg \min_{\mathcal{G}, \mathbf{A}, \{\Phi_\ell\}_{\ell=1}^L} \|E(\mathcal{G} \times_1 \mathbf{A} \times_2 \Phi_1 \times_3 \Phi_2 \times_4 \cdots \times_{L+1} \Phi_L) - \mathbf{s}\|_2^2, \quad (9.49)$$

where the elements of $\Phi_\ell \in \mathbb{C}^{N_\ell \times R_\ell}$ are $[\Phi_\ell]_{nr} = \varphi_{\ell,r}(\tau_{\ell,n})$.

As in Section 9.3.1.2, fixed-subspace reconstruction is possible with the help of training data. When it is feasible to prospectively control the $(\mathbf{k}, \tau_1, \tau_2, \dots, \tau_L)$ -space sampling in full (e.g., when all of the $\{\tau_\ell(t)\}_{\ell=1}^L$ represent sequence parameters), then one can acquire L subsets of training data, the ℓ th set of which densely samples across τ_ℓ within a limited region of $(\mathbf{k}, \{\tau_q\}_{q \neq \ell})$ -space. The SVD of each training data subset then produces each Φ_ℓ for the Tucker decomposition [22]. When $(\mathbf{k}, \tau_1, \tau_2, \dots, \tau_L)$ -space sampling cannot be fully controlled (e.g., when some of the $\{\tau_\ell(t)\}_{\ell=1}^L$ describe physiological processes such as motion), then training data may sparsely sample $(\mathbf{k}, \tau_1, \tau_2, \dots, \tau_L)$ -space with an unpredictable pattern. In such cases, Φ can instead be calculated by LRT completion of the training data followed by truncated decomposition of the training data tensor, as proposed in the MR multitasking framework [24].

ALS reconstruction can also be used to optimize the cost function in Eqs. (9.48) or (9.49) over all variables without requiring training data. However, in practice training data may already be available in the form of self-gating signals used to determine motion-related $\tau_\ell(t)$'s [24].

Implicit multidynamic low-rank tensor reconstruction

Implicit low-rank tensor modeling can be performed by penalizing a tensor rank surrogate, i.e.,

$$\hat{\mathcal{X}} = \arg \min_{\mathcal{X}} \|E(\mathcal{X}) - \mathbf{s}\|_2^2 + \lambda R_{\text{rank}}(\mathcal{X}), \quad (9.50)$$

with $R_{\text{rank}}(\mathcal{X})$ defined for example as the sum of nuclear norms for each unfolding of $\mathcal{X}$: $\sum_{i=1}^{L+1} \|\mathbf{X}_{(i)}\|_*$ [63]. Note that this approach requires storing and operating upon $\mathcal{X}$ without any parameterization; this $\mathcal{X}$ has $M \prod_{\ell=1}^L N_\ell$ complex elements. This can lead to enormous storage requirements that grow geometrically with L , making implicit low-rank tensor image reconstruction impractical for large L .

Additional multidynamic LRT models

The previous discussed extensions to low-rank modeling can be applied to multidynamic imaging as well, e.g., multidynamic LLR tensor modeling [21], multidynamic L&S [22,24], multidynamic L+S [84], spatially separable multidynamic imaging [26], multidynamic patch similarity modeling [25], and multidynamic multiscale LRT image reconstruction. Each of these strategies can support explicit or implicit low-rank reconstruction, with explicit decomposition having the aforementioned memory benefits.

9.5 Summary

Low-rank image reconstruction has enhanced or enabled several MRI applications, and active research continues into more advanced modeling, decompositions, and reconstruction algorithms. Continued development of low-rank image reconstruction for fast dynamic imaging and multidynamic imaging has the potential to change the structure of clinical MR exams from a series of independent scans to a single high-dimensional scan reconstructed as a low-rank tensor. This change would necessitate new modes of storing, viewing, and analyzing high-dimensional images, which, given the utility of low-rank decompositions for these tasks, seems a natural extension of low-rank image reconstruction to a broader low-rank image reconstruction, storage, and analysis pipeline.

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